

Jamovi Step-by-Step Lab Guide — Session 1

Course: Quantitative Data Analysis (M1 – S7) **Software:** Jamovi (Version 2.3+) **Session:** Session 1 — Data Setup & Univariate Descriptives **Target File:** data/processed/sp500_esg_jamovi.csv **Final Output:** sp500_esg_session1.omv

Step 1: Open the Dataset in Jamovi

1. Launch **Jamovi** on your computer.
 2. Click the top-left **≡ menu** (hamburger icon).
 3. Select **Open** → **Browse**.
 4. Navigate to your project folder: Group Work S&P 500/data/processed/
 5. Select **sp500_esg_jamovi.csv** (or sp500_esg_jamovi.xlsx).
 6. Click **Open**.
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Step 2: Configure Variable Measure Types (Crucial Step!)

[!CAUTION] As Dr. NGUYEN Anh-Tuan warns in **Slide 38**: *“If the measure type is wrong, Jamovi will silently refuse to offer you the right test in week 2. This is the number one cause of a lost lab session.”*

Double-click on each column header in the **Data** tab and set the **Measure type** and **Data type**:

Column Name	Measure Type in Jamovi	Data Type	Missing Values / Levels
Ticker	ID	Text	Identifier (never analysed)
Company	Nominal	Text	Company names
Sector	Nominal	Text	5 levels: <i>Consumer, Finance, Healthcare, Industrials & Energy, Tech & Comms</i>
Total_Revenue_B	Continuous	Decimal	None
Market_Cap_B	Continuous	Decimal	None
Market_Cap_Quartile	Ordinal	Text	4 ordered levels: $Q1 < Q2 < Q3 < Q4$
Profit_Margin	Continuous	Decimal	None
ROE	Continuous	Decimal	32 missing values (leave empty)
Beta	Continuous	Decimal	4 missing values
Headcount	Continuous	Integer	3 missing values
Overall_Governance_Risk	Continuous	Integer	Scores 1 to 10
Governance_Risk_Level	Ordinal	Text	3 ordered levels: <i>Low < Medium < High</i>

Step 3: Run Univariate Analysis on Categorical Variables (Slide 40)

1. Click on the **Analyses** tab in the top ribbon.
2. Click **Exploration** → **Descriptives**.
3. In the variable selection box:
 - Move **Sector** and **Governance_Risk_Level** into the **Variables** box.
4. Expand the **Statistics** panel below:
 - Check **Frequency tables** (for qualitative variables).
 - Uncheck Mean/Median/SD for qualitative variables (avoids the “average sector = 2.4” trap!).
5. Expand the **Plots** panel:
 - Check **Bar plot**.
6. **Expected Output in Jamovi:**

- Sector: Industrials & Energy (148, 29.4%), Consumer (116, 23.1%), Tech & Comms (108, 21.5%), Finance (71, 14.1%), Healthcare (60, 11.9%).
 - Governance_Risk_Level: High (197, 39.7%), Low (150, 30.2%), Medium (149, 30.0%).
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Step 4: Run Univariate Analysis & Normality Tests on Continuous Variables

1. Still in **Analyses** → **Exploration** → **Descriptives** (or create a new Descriptives block):
 2. Move the continuous variables into **Variables**:
 - Profit_Margin
 - ROE
 - Market_Cap_B
 - Total_Revenue_B
 - Overall_Governance_Risk
 3. Expand **Statistics**:
 - **Central Tendency**: Check Mean, Median, Mode.
 - **Dispersion**: Check Std. deviation, Variance, Minimum, Maximum, IQR (Interquartile range).
 - **Distribution Shape**: Check Skewness and Kurtosis.
 - **Normality**: Check **Shapiro-Wilk**.
 4. Expand **Plots**:
 - Check **Histogram**.
 - Check **Density**.
 - Check **Box plot** (and check Outliers).
 - Check **Q-Q plot**.
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Step 5: Save and Export Your .omv File (Slide 37 & 41)

1. Click the top-left ≡ **menu**.
2. Select **Save As** → **Browse**.
3. Name your file: **sp500_esg_session1.omv**.
4. Save it in your project folder. *(The .omv format bundles both your dataset and all interactive analysis output tables together into a single file ready for submission to Dr. NGUYEN Anh-Tuan).*